

Project Proposal: A Robust and Automated Simulation Pipeline for AI-Accelerated Design of Floating Offshore Wind Turbine Platforms

Project B: Programmed simulation pipeline (OpenFAST $\rightarrow$ Ansys) and dataset for ML/AIXD

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Abstract

This document proposes the development of a robust, automated, and scalable simulation pipeline to support AI-Accelerated eXtreme Design (AIXD) of floating offshore wind turbine (FOWT) platforms. The core objective is to bridge the gap between high-fidelity engineering simulation and machine learning by creating a seamless workflow that proceeds from parametric geometry generation to final structural strength assessment. The pipeline will orchestrate a series of sophisticated software tools, including Capytaine for hydrodynamics, OpenFAST for coupled time-domain analysis, and Ansys Mechanical for finite element analysis. Key innovations include the automated extraction and "staticization" of extreme load events and the curation of a clean, well-documented, and analysis-ready dataset. The final deliverable will be a production-grade pipeline and an associated data repository that will enable the rapid training and verification of AI/ML models for FOWT design optimization, significantly reducing engineering time and cost while improving design robustness.

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1 Introduction and Motivation

The design of Floating Offshore Wind Turbines (FOWTs) is a complex, multi-disciplinary challenge involving hydrodynamics, aerodynamics, structural mechanics, and control systems. Traditional design cycles are labor-intensive and time-consuming, relying on manual iterations between various simulation software packages. This process creates a significant bottleneck, hindering the rapid exploration of the vast design space required to optimize for cost, performance, and safety.

The emergence of AI-Accelerated eXtreme Design (AIXD) presents a transformative opportunity to automate and accelerate this process. However, the efficacy of any AI/ML model is fundamentally dependent on the quality, quantity, and relevance of its training data. Currently, there is a critical lack of large-scale, high-fidelity simulation datasets that are specifically curated for training such models.

This project directly addresses this gap by proposing the development of an end-to-end automated simulation pipeline. This pipeline will programmatically generate FOWT designs, perform a complete set of aero-hydro-servo-elastic analyses, identify critical load conditions, assess structural integrity, and compile the results into a structured dataset ready for consumption by AIXD frameworks. By automating these steps, we can generate the necessary data at scale, enabling a paradigm shift from manual design iterations to AI-driven optimization.

2 Project Goals and Objectives

The primary goal of this project is to implement a robust, automated, and scalable pipeline. The key objectives are to:

1. **Generate Hydrodynamics:** Programmatically generate hydrodynamic coefficients for parametric floater geometries using panel methods.
2. **Run Coupled Analyses:** Execute a large number of coupled, time-domain simulations using OpenFAST under various metocean conditions.
3. **Extract Extreme Events:** Automatically identify and extract the top N extreme loading instants from the time-series results.
4. **Map Loads to FEA:** Map the extracted dynamic loads to equivalent static load cases for Finite Element Analysis (FEA) in Ansys Mechanical.
5. **Compute Strength Metrics:** Calculate structural strength Utilization Factors (UFs) for critical components under these extreme loads.
6. **Curate a Dataset:** Compile and document a clean, version-controlled dataset suitable for training and validating AIXD models.

3 Proposed System Architecture

The pipeline will be built on a modular architecture to ensure flexibility and scalability. The high-level components are described below.

Orchestrator A Python-based workflow manager will serve as the central controller, managing job submission, dependency tracking, error handling, and data flow between components.

Geometry/Meshing • A parametric model will define the floater geometry.

- **Capytaine** will be used for automated panel mesh generation for hydrodynamic analysis.

- A parametric FEA mesh generator, using either Ansys Mechanical scripting or PyMAPDL, will create the structural mesh.

Solvers • **Hydrodynamics:** `Capytaine` will solve the radiation/diffraction problem to generate the hydrodynamic database.

- **Coupled Dynamics:** `OpenFAST` will perform the time-domain coupled simulations of the complete FOWT system.
- **Structural Strength:** `Ansys Mechanical` will be used for static strength checks with inertial relief to account for dynamic effects.

Data Layer • A structured results store will use efficient formats like `Parquet/Arrow` for tabular data and `HDF5` for time-series data.

- Large file versioning will be managed with `DVC` or `git-LFS`, and all configurations will be semantically versioned.

ML Interface • The project will define dataset schemas and provide data loaders for seamless integration with the AIXD framework.

- The dataset will be pre-split into training, validation, and test sets, accompanied by a comprehensive metadata registry.

[A block diagram illustrating the data flow and component interactions will be included in the final design document.]

4 Methodology and Work Packages

The project will be executed through a series of well-defined work packages (WPs).

4.1 WP1: Configuration and Parameterization

This foundational step involves creating a centralized configuration system. A single `YAML` file per design case will define all parameters, including:

- **Design Parameters:** Floater geometry, material properties, mooring system layout.
- **Environmental Conditions:** Site-specific metocean bins (wind speeds, wave states, etc.).
- **Numerical Controls:** Simulation time steps, random seeds for turbulence and waves (ensuring reproducibility), mesh density settings, and solver convergence criteria.
- **Extraction Policy:** A clear definition for identifying the N extreme instants, such as Peaks-Over-Threshold (POT) criteria and separation windows to avoid clustering.

4.2 WP2: Hydrodynamics Generation

For each parametric design, the pipeline will automatically perform a frequency-domain hydrodynamic analysis.

1. Generate a watertight panel mesh from the parametric geometry, ensuring it meets specified quality metrics.
2. Execute `Capytaine` to compute frequency-dependent hydrodynamic coefficients (added mass, radiation damping) and wave excitation forces.
3. Export the results into OpenFAST-compatible formats (`.1`, `.3`, `.hst`).

4. Store summary descriptors (e.g., added mass at specific frequencies, hydrostatic stiffness matrix) as features for the ML dataset.

4.3 WP3: Coupled OpenFAST Simulations

This WP focuses on running the core time-domain simulations for a defined set of Design Load Cases (DLCs), such as DLC 1.2 (power production with normal turbulence).

1. For each case, generate turbulent wind fields (**TurbSim**) and irregular wave time series (**HydroDyn**) using a fixed list of seeds.
2. Run the full OpenFAST simulation, capturing time histories of key outputs: platform motions, tower base loads, nacelle accelerations, and mooring line tensions.
3. Implement automated Quality Control (QC) gates to validate simulation integrity, including checks for NaNs/inf, energy balance, and excessive constraint violations in the model.

4.4 WP4: Extreme Event Extraction and Staticization

This critical step translates dynamic simulation results into a format suitable for static structural analysis.

1. **Identify Instants:** Scan time-series data to find candidate extreme events based on the pre-defined policy (e.g., peaks in tower base bending moments, platform pitch, or mooring tension). Enforce a time separation window to ensure distinct events are selected.
2. **Build Static Load Set:** For each selected instant, assemble an equivalent static load set that includes:
 - The 6-DOF resultant force and moment vector at the tower base.
 - Tension vectors for each mooring line applied at the fairlead nodes.
 - The hydrostatic pressure field corresponding to the platform's instantaneous draft, roll, and pitch.
3. **Harmonize and Balance:** Implement robust procedures for coordinate system transformations and unit conversions between OpenFAST and Ansys. Apply inertial relief in Ansys to balance the external loads and account for the body's own inertia, approximating the missing dynamic effects.

4.5 WP5: Ansys Mechanical Strength Check

The staticized load sets will be used to perform FEA-based strength checks.

1. Programmatically mesh the structure in Ansys using shell elements appropriate for thin-walled marine structures.
2. Apply each static load set and solve the linear static analysis, including inertial relief.
3. Post-process the results to compute the Utilization Factor (UF) for pre-selected structural components, defined as:

$$UF = \frac{\sigma_{eq}}{\sigma_{allow}}$$

where σ_{eq} is the equivalent stress (e.g., von Mises) and σ_{allow} is the allowable stress (material yield strength divided by a partial safety factor).

4. Store the maximum UF for each component by enveloping the results from all N extreme cases.

4.6 WP6: Dataset Compilation for ML

The final step is to collate all generated data into a structured, analysis-ready format. The dataset for each design will contain several feature blocks:

- **Design Features:** Parametric inputs for geometry, materials, and mooring.
- **Hydrodynamic Features:** Condensed descriptors from the Capytaine analysis.
- **Environmental Features:** Metocean parameters for the simulation set.
- **Response Features:** Statistical summaries of the OpenFAST time-series (RMS, spectral peaks, etc.) and the context of each selected extreme instant.
- **Target Labels:** The calculated maximum Utilization Factors (UFs) for each component.

4.7 WP7: AIXD Integration and Verification Loop

The curated dataset will be integrated with the AIXD framework to enable a complete design-analysis-optimization loop.

1. Develop dataset adapters and loaders to feed the data into AIXD models for two primary tasks:
 - **Forward Prediction:** Mapping design parameters to UF targets.
 - **Optimization:** Finding optimal designs that minimize mass while satisfying constraints (e.g., $UF \leq 1.0$).
2. Implement a verification loop where candidate designs generated by AIXD are fed back into this pipeline to generate "ground truth" simulation results. This allows for continuous model validation and refinement.

4.8 WP8: Results Verification

This work package is dedicated to the independent verification of the simulation results by the hydrodynamics team (Team 2). Following the completion of the Ansys strength checks, this team will:

1. **Review Final Outputs:** Analyze the final Utilization Factors (UFs) and the corresponding extreme event load sets from WP4 and WP5.
2. **Correlate with Hydrodynamic Inputs:** Cross-reference the conditions of extreme events with the initial hydrodynamic database generated in WP2. This ensures that the final structural hotspots are driven by physically consistent hydrodynamic phenomena.
3. **Provide Feedback:** Document any discrepancies or unexpected behaviors, providing critical feedback to the automation team (Team 1) to refine the pipeline's fidelity and Quality Control (QC) gates.

This step serves as a crucial quality assurance loop, ensuring that the data generated for the ML model is not only automatically produced but also rigorously vetted by domain experts.

5 Project Deliverables

Upon completion, this project will deliver:

1. A fully operational, documented, and version-controlled Python-based simulation pipeline.
2. A comprehensive, curated dataset of FOWT simulations, stored in Parquet and HDF5 formats, complete with a data schema and metadata registry.
3. A set of data loaders and adapters for interfacing the dataset with the AIXD framework.
4. A verification report detailing the performance of the pipeline and providing initial results from the AIXD verification loop.
5. All source code, configuration files, and scripts developed during the project.

6 Proposed Timeline

The project is scheduled to run from October 3rd to December 20th. The Gantt chart below (Figure 1) visualizes the allocation of work packages across this timeline.

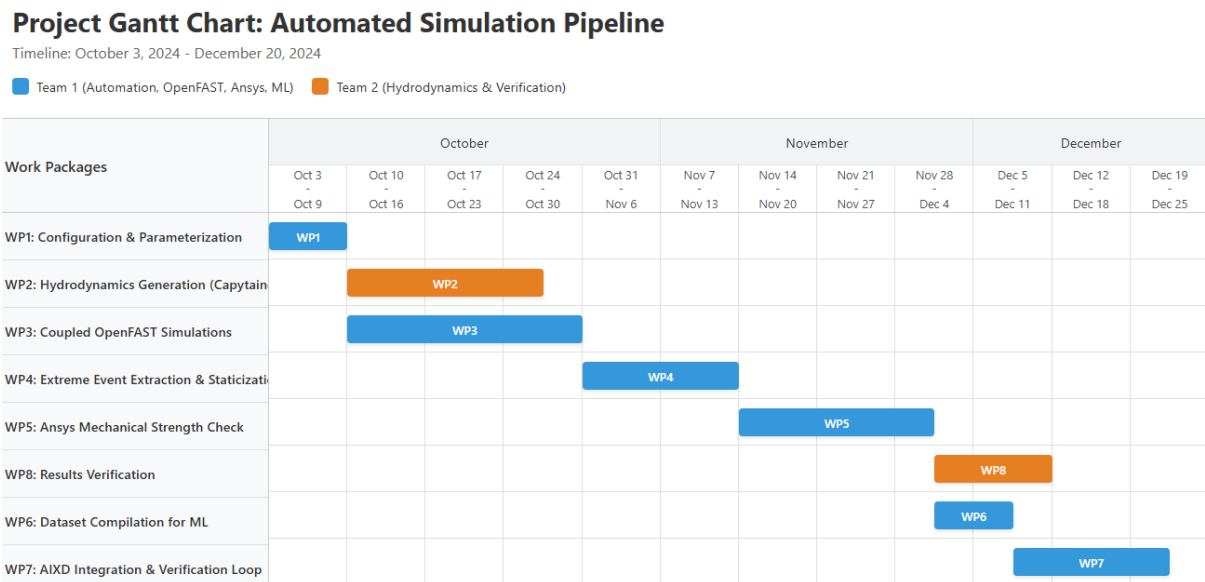


Figure 1: Gantt chart for the project timeline (Oct 3 - Dec 20). Team 1 (blue) is responsible for the main automation pipeline and ML integration. Team 2 (orange) handles hydrodynamics generation and final results verification.

7 Required Resources

- Software**
- Python 3.x with standard scientific libraries (NumPy, Pandas, SciPy).
 - Commercial and open-source solvers: Ansys Mechanical, OpenFAST, Capytaine.
 - Workflow management and data versioning tools: DVC/git-LFS.
- Hardware**
- Development workstations for pipeline creation and testing.
 - Access to a High-Performance Computing (HPC) cluster for running the large-scale simulation campaigns in parallel.

- Personnel**
- Project lead with expertise in simulation workflows and software development.
 - Engineering specialists with knowledge of hydrodynamics, structural mechanics, and offshore wind systems.